

Type-B Self-Similarity of the Fake-Degree Frobenius Sum at a Root of Unity: The Odd- e Case

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Abstract

Let $W_B = \mathbb{Z}/2 \wr S_n$ be the hyperoctahedral group with reflection representation $V \cong \mathbb{C}^n$ and degrees $\{2, 4, \dots, 2n\}$. For a signed cycle type (α, β) (with $|\alpha| + |\beta| = n$) put

$$\Psi_{(\alpha, \beta)}^B(t) := \frac{\prod_{i=1}^n (1 - t^{2i})}{\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)} \in \mathbb{Z}[t],$$

$z_{(\alpha, \beta)}^B := 2^{\ell(\alpha) + \ell(\beta)} z_\alpha z_\beta$, and, for ζ_e a primitive e -th root of unity,

$$q_e^{B(n)} := \sum_{|\alpha| + |\beta| = n} \frac{\Psi_{(\alpha, \beta)}^B(\zeta_e)}{z_{(\alpha, \beta)}^B} p_\alpha(x) p_\beta(y) \in \Lambda(x) \otimes \Lambda(y) \otimes \mathbb{Z}[\zeta_e].$$

Theorem. For every odd $e \geq 3$ and every $n \geq 0$, with $k := \lfloor n/e \rfloor$ and $r_0 := n \bmod e$,

$$\boxed{q_e^{B(n)} = p_e(x)^k \cdot q_e^{B(r_0)}.$$

The proof is a mechanical translation of the type-A self-similarity theorem [2]: for odd e , the negative-cycle bipartition β enters every coefficient as an inert multiplicative constant $\prod_b (1 + \zeta_e^b) \in \mathbb{Z}[\zeta_e]^\times$, and the positive-side analysis is identical to type A after a single reindexing $i \mapsto 2i$ (which is invertible mod e for odd e). The k -shift on the positive side cancels via the same identity $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = e$ that closed the type-A theorem.

The even- e case is substantively different — it requires a strictly stronger, two-sided support theorem plus a companion identity $\Psi_{\emptyset, (e/2, e/2)}^B(\zeta_e) = 2e^2$ — and is deferred to a follow-up.

1 Setup

Fix odd $e \geq 3$ and let $\zeta_e := e^{2\pi i/e}$. Set $\alpha_r := 1 - \zeta_e^r$ for $r \in \{1, \dots, e-1\}$; each α_r is a nonzero element of $\mathbb{Z}[\zeta_e]$. Throughout, $n \geq 0$, $k := \lfloor n/e \rfloor$, and $r_0 := n \bmod e \in \{0, 1, \dots, e-1\}$.

Bipartitions of n index both the conjugacy classes and the irreducible characters of W_B . For a class of signed cycle type (α, β) with $|\alpha| + |\beta| = n$, the characteristic polynomial of a representative $w \in W_B$ acting on V is

$$\det(1 - tw|_V) = \prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b),$$

and by Chevalley–Molien for the reflection group W_B ([5, §3.20], cf. [3]) the graded character of the W_B -coinvariant algebra at w is

$$\Psi_{(\alpha, \beta)}^B(t) = \frac{\prod_{i=1}^n (1 - t^{2i})}{\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)}, \tag{1}$$

a polynomial in $\mathbb{Z}[t]$. The centraliser $|Z_{W_B}(w)| = z_{(\alpha,\beta)}^B := 2^{\ell(\alpha)+\ell(\beta)} z_\alpha z_\beta$, where $z_\gamma := \prod_i i^{m_i(\gamma)} m_i(\gamma)!$.

Writing $q_e^{B(n)} = \sum_{(\alpha,\beta)} c_{\alpha,\beta}^{B(n)} p_\alpha(x) p_\beta(y)$, the coefficients are

$$c_{\alpha,\beta}^{B(n)} = \frac{\Psi_{(\alpha,\beta)}^B(\zeta_e)}{z_{(\alpha,\beta)}^B}. \quad (2)$$

2 Cyclotomic and Arithmetic Lemmas

Lemma 1 (Classical cyclotomic identity). $\prod_{r=1}^{e-1} \alpha_r = \prod_{r=1}^{e-1} (1 - \zeta_e^r) = e$.

Proof. Setting $x = 1$ in the factorisation $x^e - 1 = (x - 1) \prod_{r=1}^{e-1} (x - \zeta_e^r)$ and taking the limit gives $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = \lim_{x \rightarrow 1} (x^e - 1)/(x - 1) = e$. $\square$

Lemma 2 (Numerator: multiplicity of e -cycle zeros). *For odd e , $\prod_{i=1}^n (1 - t^{2i})$ has a zero of order exactly k at $t = \zeta_e$, contributed by the k indices $i \in \{e, 2e, \dots, ke\}$. Moreover*

$$\lim_{t \rightarrow \zeta_e} \frac{\prod_{i=1}^k (1 - t^{2ei})}{(1 - t^e)^k} = 2^k k!.$$

Proof. For odd e , $\gcd(2, e) = 1$, so $e \mid 2i \iff e \mid i$; the numerator factor $1 - \zeta_e^{2i}$ vanishes iff $e \mid i$, giving exactly k simple zeros. For $i = em$, write $1 - t^{2em} = 1 - (t^e)^{2m}$; substituting $u := t^e$,

$$\lim_{t \rightarrow \zeta_e} \prod_{m=1}^k \frac{1 - t^{2em}}{1 - t^e} = \lim_{u \rightarrow 1} \prod_{m=1}^k \frac{1 - u^{2m}}{1 - u} = \prod_{m=1}^k 2m = 2^k k!,$$

using $\lim_{u \rightarrow 1} (1 - u^d)/(1 - u) = d$ termwise. $\square$

Lemma 3 (Numerator: non- e -cycle residues). *For odd e and $n = ke + r_0$ with $0 \leq r_0 < e$,*

$$\prod_{\substack{1 \leq i \leq n \\ e \nmid i}} (1 - \zeta_e^{2i}) = e^k \prod_{s=1}^{r_0} (1 - \zeta_e^{2s}).$$

Proof. The map $\phi(i) := 2i \bmod e$ is a bijection on the residue classes $\{1, 2, \dots, e-1\}$ because $\gcd(2, e) = 1$. As i runs over $[1, n]$ with $e \nmid i$, its residue $i \bmod e$ takes each value in $\{1, \dots, e-1\}$ exactly k times (from the k full cycles $[1, e], [e+1, 2e], \dots, [(k-1)e+1, ke]$) plus one extra copy of each value in $\{1, 2, \dots, r_0\}$ (from $i \in [ke+1, ke+r_0]$). Applying ϕ , the residue $2i \bmod e$ takes each value in $\{1, \dots, e-1\}$ exactly k times, plus one extra copy of each value in $\{2s \bmod e : s = 1, \dots, r_0\}$. Since ζ_e^{2i} depends only on $2i \bmod e$,

$$\prod_{\substack{i \in [1, n] \\ e \nmid i}} (1 - \zeta_e^{2i}) = \left(\prod_{r=1}^{e-1} \alpha_r \right)^k \prod_{s=1}^{r_0} (1 - \zeta_e^{2s \bmod e}).$$

Now $\prod_{r=1}^{e-1} \alpha_r = e$ by Lemma 1, and $\zeta_e^{2s \bmod e} = \zeta_e^{2s}$ (since $\zeta_e^e = 1$). This gives the claim. $\square$

Lemma 4 (Negative-cycle factor is a unit). *For odd e and every partition β , $N_\beta(\zeta_e) := \prod_{b \in \beta} (1 + \zeta_e^b) \in \mathbb{Z}[\zeta_e]^\times$.*

Proof. $1 + \zeta_e^b = 0 \iff \zeta_e^b = -1$; since $-1 = \zeta_e^{e/2}$ requires $2 \mid e$, no such b exists for odd e . Each factor is nonzero, so the product is a nonzero element of the Dedekind domain $\mathbb{Z}[\zeta_e]$. To see it is a unit: $1 + \zeta_e^b$ divides $1 - \zeta_e^{2b}$, and the latter is a unit in $\mathbb{Z}[\zeta_e]$ whenever $\gcd(2b, e) = \gcd(b, e)$ (odd e) equals $\gcd(b, e)$, which is a divisor of e ; when $e \nmid b$, $1 - \zeta_e^{2b}$ has ideal norm dividing that of $1 - \zeta_e^{\gcd(2b, e)}$, and each of these has finite norm dividing $e^{[Q(\zeta_e):\mathbb{Q}]}$, so is a unit up to a factor lying above e . A cleaner route: over $\mathbb{Z}[\zeta_e]$, the only prime that can divide $1 + \zeta_e^b$ is the unique prime above e ; but $1 + \zeta_e^b \equiv 1 + 1 = 2 \pmod{(1 - \zeta_e)}$, and 2 is a unit modulo the prime above e (since $\gcd(2, e) = 1$). Hence $1 + \zeta_e^b$ is a unit.¹ $\square$

3 The Coefficient Formula

Let $\alpha = (e^k, \alpha')$ where α' is a partition all of whose parts are $< e$; in particular α' has no parts divisible by e , and $|\alpha'| = |\alpha| - ek$. Let β be any partition, and set $m_r(\alpha') :=$ the multiplicity of r in α' for $r \in \{1, \dots, e-1\}$.

Proposition 5 (Closed-form coefficient). *For odd e , all $n \geq 0$, and every pair (α, β) with $\alpha = (e^k, \alpha')$ (all parts of α' strictly less than e) and $|\alpha| + |\beta| = n$,*

$$c_{(e^k, \alpha'), \beta}^{B, (n)} = \frac{1}{z_{(\alpha', \beta)}^B N_\beta(\zeta_e)} \prod_{r=1}^{e-1} \alpha_r^{g_r(r_0) - m_r(\alpha')}, \quad (3)$$

where $g_r(r_0) := |\{s \in \{1, \dots, r_0\} : 2s \equiv r \pmod{e}\}| \in \{0, 1\}$ and $|\alpha'| + |\beta| = r_0 := n \bmod e$.

Proof. Split the numerator of $\Psi_{(\alpha, \beta)}^B$ by whether $e \mid i$:

$$\prod_{i=1}^n (1 - t^{2i}) = \left(\prod_{m=1}^k (1 - t^{2em}) \right) \prod_{\substack{i \in [1, n] \\ e \nmid i}} (1 - t^{2i}).$$

The positive-cycle denominator splits as $\prod_a (1 - t^a) = (1 - t^e)^k \prod_{a' \in \alpha'} (1 - t^{a'})$ because α' has no parts divisible by e . Substitute and take $t \rightarrow \zeta_e$:

$$\Psi_{(\alpha, \beta)}^B(\zeta_e) = \left(\lim_{t \rightarrow \zeta_e} \frac{\prod_{m=1}^k (1 - t^{2em})}{(1 - t^e)^k} \right) \cdot \frac{\prod_{i: e \nmid i} (1 - \zeta_e^{2i})}{\prod_{a' \in \alpha'} (1 - \zeta_e^{a'}) N_\beta(\zeta_e)}.$$

The remaining factors are all finite: the parts of α' lie in $\{1, \dots, e-1\}$, so each $1 - \zeta_e^{a'} = \alpha_{a'} \neq 0$; and $N_\beta(\zeta_e) \neq 0$ by Lemma 4. By Lemmas 2 and 3,

$$\Psi_{(\alpha, \beta)}^B(\zeta_e) = 2^k k! \cdot e^k \cdot \frac{\prod_{s=1}^{r_0} (1 - \zeta_e^{2s})}{\prod_{r=1}^{e-1} \alpha_r^{m_r(\alpha')} \cdot N_\beta(\zeta_e)}. \quad (4)$$

Grouping the extra copies of ζ_e^{2s} ($s = 1, \dots, r_0$) by residue class,

$$\prod_{s=1}^{r_0} (1 - \zeta_e^{2s}) = \prod_{r=1}^{e-1} \alpha_r^{g_r(r_0)},$$

¹We only need that $N_\beta(\zeta_e)$ is a nonzero element of $\mathbb{Z}[\zeta_e]$ (well-defined denominator in an evaluation of $\Psi^B(\zeta_e)$). Unithood is stronger than we need; the reader may verify it for concreteness.

so

$$\Psi_{(\alpha,\beta)}^B(\zeta_e) = 2^k k! e^k \cdot \frac{\prod_{r=1}^{e-1} \alpha_r^{g_r(r_0) - m_r(\alpha')}}{N_\beta(\zeta_e)}. \quad (5)$$

Since $\alpha = (e^k, \alpha')$ and α' has no parts equal to e , we have $z_\alpha = e^k k! z_{\alpha'}$, and $\ell(\alpha) = k + \ell(\alpha')$, so

$$z_{(\alpha,\beta)}^B = 2^{k+\ell(\alpha')+\ell(\beta)} e^k k! z_{\alpha'} z_\beta = 2^k k! e^k \cdot z_{(\alpha',\beta)}^B. \quad (6)$$

Substituting (5) and (6) into $c_{(e^k, \alpha'), \beta}^{B, (n)} = \Psi_{(\alpha,\beta)}^B(\zeta_e) / z_{(\alpha,\beta)}^B$ cancels the factor $2^k k! e^k$ and yields (3). $\square$

4 Support Theorem

Theorem 6 (Support of $q_e^{B, (n)}$, odd e). *For odd $e \geq 3$ and $n \geq 0$, $c_{\alpha, \beta}^{B, (n)} \neq 0$ if and only if*

$$\alpha = (e^k, \alpha'), \quad \text{all parts of } \alpha' \text{ strictly less than } e, \quad |\alpha'| + |\beta| = r_0.$$

Proof. Since $\Psi_{(\alpha,\beta)}^B(t) \in \mathbb{Z}[t]$ (Chevalley–Molien), $\Psi_{(\alpha,\beta)}^B(\zeta_e) = 0$ if and only if the naive rational-function evaluation of (1) has a strictly higher-order zero in the numerator than in the denominator at $t = \zeta_e$. Let ord_{ζ_e} denote order of vanishing.

Numerator: $1 - \zeta_e^{2i} = 0 \iff e \mid 2i \iff e \mid i$ (odd e), so $\text{ord}_{\zeta_e} \prod_{i=1}^n (1 - t^{2i}) = k$.

Denominator: $1 - \zeta_e^a = 0 \iff e \mid a$, so $\text{ord}_{\zeta_e} \prod_a (1 - t^a) = |\{a \in \alpha : e \mid a\}| =: k_e(\alpha)$. And $\text{ord}_{\zeta_e} \prod_b (1 + t^b) = |\{b \in \beta : \zeta_e^b = -1\}| = 0$ for odd e .

Hence $\Psi_{(\alpha,\beta)}^B(\zeta_e) \neq 0 \iff k_e(\alpha) \geq k$. Since $k_e(\alpha) \cdot e \leq |\alpha| \leq n < (k+1)e$, we have $k_e(\alpha) \leq k$; combined we need $k_e(\alpha) = k$.

Now $k_e(\alpha) = k$ means α has exactly k parts divisible by e ; write those parts as $ea_1, \dots, ea_k$ with each $a_j \geq 1$. Their sum is $e \sum_j a_j \leq |\alpha| \leq n = ke + r_0 < (k+1)e$, so $\sum_j a_j \leq k$; combined with $\sum_j a_j \geq k$, we get $a_j = 1$ for all j . Thus $\alpha = (e^k, \alpha')$ where the remaining parts α' have no part divisible by e . Then $|\alpha'| + |\beta| = n - ek = r_0 < e$, so all parts of α' are $\leq r_0 < e$.

Conversely, if $\alpha = (e^k, \alpha')$ with parts of α' in $\{1, \dots, e-1\}$: then $k_e(\alpha) = k$, the pole/zero balance is even, and the closed form (3) shows $c_{\alpha, \beta}^{B, (n)}$ is a nonzero element of $\mathbb{Z}[\zeta_e]$ (numerator a nonzero product of α_r 's in $\mathbb{Z}[\zeta_e]^\times$ up to units, denominator $z^B \cdot N_\beta(\zeta_e)$ nonzero by Lemma 4). $\square$

5 Main Theorem

Theorem 7 (Type-B self-similarity, odd e). *For every odd $e \geq 3$ and every $n \geq 0$, with $k = \lfloor n/e \rfloor$ and $r_0 = n \bmod e$,*

$$q_e^{B, (n)} = p_e(x)^k \cdot q_e^{B, (r_0)}.$$

Equivalently, $p_e^B := p_{(e)}(x)$, and the “residue factor” $F_e^{B, (n)} := q_e^{B, (n)} / p_e(x)^k$ satisfies $F_e^{B, (n)} = q_e^{B, (r_0)}$; in particular it depends on n only through r_0 .

Proof. Apply Proposition 5 both at parameter n and at parameter r_0 (where $\lfloor r_0/e \rfloor = 0$ since $0 \leq r_0 < e$).

At parameter r_0 : for every pair (α', β) with $|\alpha'| + |\beta| = r_0$, all parts of α' are automatically $\leq r_0 < e$, so $\alpha' = (e^0, \alpha')$ is the trivial decomposition and (3) reads

$$c_{\alpha', \beta}^{B, (r_0)} = \frac{1}{z_{(\alpha', \beta)}^B N_\beta(\zeta_e)} \prod_{r=1}^{e-1} \alpha_r^{g_r(r_0) - m_r(\alpha')}. \quad (7)$$

By the support theorem 6 applied at r_0 (with $k = 0$), the support of $q_e^{B(r_0)}$ is exactly $\{(\alpha', \beta) : |\alpha'| + |\beta| = r_0\}$ (all such pairs contribute).

At parameter n : by the support theorem 6, the support of $q_e^{B(n)}$ is $\{((e^k, \alpha'), \beta) : |\alpha'| + |\beta| = r_0, \alpha' \text{ has parts } < e\}$. The parts-condition is automatic since $|\alpha'| \leq r_0 < e$. Comparing (3) and (7) we obtain, for every such (α', β) ,

$$c_{(e^k, \alpha'), \beta}^{B(n)} = c_{\alpha', \beta}^{B(r_0)}. \quad (8)$$

Assemble:

$$\begin{aligned} q_e^{B(n)} &= \sum_{|\alpha'| + |\beta| = r_0} c_{(e^k, \alpha'), \beta}^{B(n)} p_{(e^k, \alpha')}(x) p_\beta(y) && \text{(support at } n) \\ &= \sum_{|\alpha'| + |\beta| = r_0} c_{(e^k, \alpha'), \beta}^{B(n)} p_e(x)^k p_{\alpha'}(x) p_\beta(y) && (p_{(e^k, \alpha')} = p_e^k p_{\alpha'}) \\ &= p_e(x)^k \sum_{|\alpha'| + |\beta| = r_0} c_{\alpha', \beta}^{B(r_0)} p_{\alpha'}(x) p_\beta(y) && \text{(by (8))} \\ &= p_e(x)^k q_e^{B(r_0)}. && \text{(support at } r_0) \end{aligned}$$

The last step uses that the full sum over $\{(\alpha', \beta) : |\alpha'| + |\beta| = r_0\}$ is $q_e^{B(r_0)}$, which is the support of $q_e^{B(r_0)}$. $\square$

6 Corollaries

Corollary 8 (n -independence of the type-B residue factor). $F_e^{B(n)} := q_e^{B(n)} / p_e(x)^{\lfloor n/e \rfloor}$ depends on n only through $r_0 = n \bmod e$; explicitly,

$$F_e^{B(n)} = q_e^{B(r_0)}.$$

Proof. Immediate from Theorem 7. $\square$

Corollary 9 (Multiplicative recursion). For every odd $e \geq 3$ and every $n \geq e$,

$$q_e^{B(n)} = p_e(x) \cdot q_e^{B(n-e)}.$$

Equivalently, $q_e^{B(qe+r)} = p_e(x)^q q_e^{B(r)}$ for $q \geq 0$ and $0 \leq r < e$.

Proof. Write $n = qe + r_0$ with $0 \leq r_0 < e$ and apply Theorem 7 twice: to $q_e^{B(n)}$ (giving $p_e^q q_e^{B(r_0)}$) and to $q_e^{B(n-e)} = q_e^{B((q-1)e+r_0)}$ (giving $p_e^{q-1} q_e^{B(r_0)}$). $\square$

Corollary 10 (Fundamental domain, odd e). For every odd $e \geq 3$, the sequence $(q_e^{B(n)})_{n \geq 0}$ is the $p_e(x)$ -linear extension of its e “basic” values

$$q_e^{B(0)}, q_e^{B(1)}, \dots, q_e^{B(e-1)}.$$

Each $q_e^{B(r)}$ ($0 \leq r < e$) is a symmetric-function-in-two-alphabets of bidegree (a, b) summed over $a + b = r$, with power-sum coefficients in $\mathbb{Z}[\zeta_e]$ given by (7).

Proof. Direct from Corollary 9. $\square$

Remark 11 (Sanity check: $\beta = \emptyset$ reduction). When $\beta = \emptyset$, $N_\emptyset(\zeta_e) = 1$, and the formula (3) reads

$$c_{(e^k, \alpha'), \emptyset}^{B, (n)} = \frac{1}{z_{(\alpha', \emptyset)}^B} \prod_{r=1}^{e-1} \alpha_r^{g_r(r_0) - m_r(\alpha')}.$$

Compared with the type-A byproduct [1] for $\alpha = (e^k, \alpha')$,

$$c_{(e^k, \alpha')}^{A, (n)} = \frac{1}{e^k z_{\alpha'}} \prod_{r=1}^{e-1} \alpha_r^{k+1[r \leq r_0] - m_r(\alpha')},$$

one sees two differences:

- a factor $2^{\ell(\alpha')}$ in $z_{(\alpha', \emptyset)}^B$ versus $z_{\alpha'}$: this is the type-B centraliser correction and reflects that $p_\alpha(x) \in \Lambda(x)$ contains only positive cycles;
- the type-A “extra exponent” $\mathbf{1}[r \leq r_0]$ becomes $g_r(r_0) = \mathbf{1}[r \in \phi([1, r_0])]$ in type B, where $\phi(s) = 2s \bmod e$. Both formulas encode the same r_0 “extra copies” of numerator residues; type A uses $\{i : ke < i \leq n\}$ with residues $\{1, \dots, r_0\}$, type B uses $\{2i : ke < i \leq n\}$ with residues $\{\phi(1), \dots, \phi(r_0)\}$.

Both agree with the general principle that the coefficient is determined by the residue-class multiset of the “leftover” cycles.

Remark 12 (Comparison with the Verschiebung route). The 2026-08-06 dream projected Ayer–Kumari’s odd-power-of-even-root factorisation [4] as the “right” tool for type-B self-similarity (as the classical-groups analogue of Albion’s Verschiebung theorem). Theorem 7 bypasses that machinery: the collapse is a one-line application of the cyclotomic identity in Lemma 1, in the same spirit as the type-A support and self-similarity theorems [1, 2]. This is the third instance in the 2026-08 sprint of a Chevalley–Molien plus classical-cyclotomy argument sufficing where a modern 2025 tool was projected as necessary. For the odd- e case, the mechanism is transparent: for odd e , the “second alphabet” β is entirely inert (Lemma 4), and the positive-cycle analysis is the type-A calculation transported through the reindexing $i \mapsto 2i$.

Remark 13 (Why even e is different). For even e , three features of the odd- e argument break simultaneously:

1. the reindexing $i \mapsto 2i \bmod e$ is 2-to-1 rather than 1-to-1;
2. the negative-cycle factor $\prod_b (1 + \zeta_e^b)$ has zeros (namely when $b \in (e/2) \cdot \mathbb{Z}$), so the negative side is not inert;
3. the numerator has “too many” zeros ($\lfloor 2n/e \rfloor$ rather than $\lfloor n/e \rfloor$).

The compute-based discovery report [3] identifies a natural even- e analogue: the residue factor absorbs $p_e^B := p_{(e/2, e/2)}(y)$ (living entirely on the negative side), together with a strictly stronger, two-sided support theorem and a companion cyclotomic identity $\Psi_{\emptyset, (e/2, e/2)}^B(\zeta_e) = 2e^2$. The full even- e theorem is a follow-up.

7 Verification

The identity $q_e^{B(n)} = p_e(x)^k q_e^{B(r_0)}$ was independently verified with exact arithmetic in $\mathbb{Z}[\zeta_e]$ (no floating point) at all odd- e test pairs of [3, ?]:

$$(n, e) \in \{(5, 3), (6, 3), (7, 3), (8, 3), (9, 3), (11, 3); (7, 5), (9, 5), (11, 5), (13, 5)\}.$$

Residuals are *exactly zero* on every test pair. The verification script computes $\Psi_{(\alpha, \beta)}^B(t)$ as a polynomial in $\mathbb{Z}[t]$ via exact long division of $\prod(1 - t^{2i})$ by $\prod(1 - t^a) \prod(1 + t^b)$, evaluates at the cyclotomic integer $\zeta_e \in \mathbb{Z}[\zeta_e]$ via reduction modulo the minimal polynomial $\Phi_e(t)$, and directly compares the bipartition power-sum coefficients on both sides of the theorem.

As a supplementary check, the closed form (3) predicts, for $(n, e) = (5, 3)$ (so $k = 1$, $r_0 = 2$) and $(\alpha, \beta) = ((3, 2), \emptyset)$:

$$c_{(3,2),\emptyset}^{B,(5)} = \frac{1}{z_{(2),\emptyset}^B} \alpha_1^{g_1(2)-0} \alpha_2^{g_2(2)-1} = \frac{1}{4} \alpha_1^1 \alpha_2^0 = \frac{1 - \zeta_3}{4},$$

using $g_1(2) = 1$ (from $s = 2: 2 \cdot 2 \bmod 3 = 1$), $g_2(2) = 1$ (from $s = 1$), $m_2((2)) = 1$, $m_1((2)) = 0$, and $z_{(2),\emptyset}^B = 2^{1+0} \cdot z_{(2)} \cdot z_{\emptyset} = 2 \cdot 2 \cdot 1 = 4$. This matches $c_{(2),\emptyset}^{B,(2)}$ computed at parameter $r_0 = 2$: $\Psi_{(2),\emptyset}^B(t)|_{t=\zeta_3} = (1 - t^2)(1 - t^4)/(1 - t^2)|_{t=\zeta_3} = 1 - \zeta_3$, divided by $z_{(2),\emptyset}^B = 4$.

Acknowledgements

This proof is the third theorem in the 2026-08 sprint. The type-A support theorem [1] and type-A self-similarity theorem [2] both closed via the same Chevalley–Molien + classical-cyclotomy mechanism; the odd- e type-B statement transports mechanically through the reindexing $i \mapsto 2i$. The even- e case remains open.

References

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